Calculatoare numerice

(lab 6 - S6)

02.04.2025

BRÎNDA Sefu' meu (nu-i adevărat)

BRÎNDA E SEXY (aşa zice el)

II. Processor performance

→ Execution time = $\frac{1}{\text{Performance}}$

User execution time

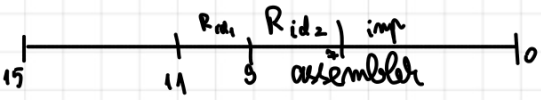
CPU time

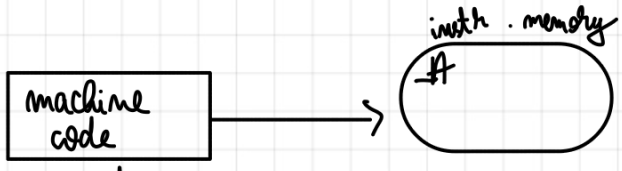
C++ ↓ compiler

instructions

e.g. (ADD R_1, R_2, R_3)

instructions set architecture





addresses mode

R-type = register type

→ primii 5 biți → OP codes → câte operații

i-type

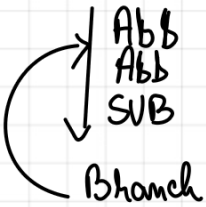
M-type

→ ALU

→ MEM

→ branch / flow control

PC program contor = 0



Throughput (programe / unitate de timp)

CPU_{time} (s) = clock cycle time (CCT) × clock cycles (CC) s/clock

$S = \frac{S}{clock}$ clock $\frac{CC}{CR}$ $CR = \frac{1}{CCT}$

CC = clocks per instr (CPi) × instruction count (ic)

$CPU_{time} = ic \times CPi \times CCT = \frac{ic \times CPi}{CR}$ -> processor perf.

RISC CISC

ISA

1.

INSR type

	A	B	C
CPi	1	2	3

	A	B	C
CS_1	2	1	2
CS_2	4	1	1

a) Which CS executes more instructions?

b) Which is faster?

c) What is their CPi?

a) $ic_{CS_1} = 2+1+2 = 5$

$ic_{CS_2} = 4+1+1 = 6$ CS2

b) $CC = \sum_{i=1}^{n} ic \times CPi \cdot i \Rightarrow CC_{CS1} = 2\times1 + 1\times2 + 2\times3 = 10$

$CC_{CS2} = 4\times1 + 1\times2 + 1\times3 = 9$

c) $CPi_{CS_1} = \frac{CC_{CS1}}{ic_{CS1}} = \frac{10}{5} = 2$ average CPi

$CPi_{CS2} = \frac{9}{6} = 1,5$

1.6.

	CPi				
	A	B	C	D	CR
P1	1	2	3	3	2,5 GHz
P2	2	2	2	2	3,6 GHz

IC = 1.0 E6 | 10% A
20% B
50% C
20% D

P_1: ICC = 0.1 × IC + 0.2 · IC × 2 + 0.5 × IC × 3 + 0.2 × IC × 3 = $2.6 \cdot 10^6$

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